

Self assessment report for Facto-DSA NGCC Public Key Cryptographic Algorithm Call

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June 2026

By following the self-assessment guidelines seen in the previous section, we execute `ngcc.bench` for each target level bit security to measure the Facto-DSA implementation versions (Reference Implementation, Performance Optimization Version and Resource Optimization Version). These are included in each test report JSON.

These indicators are collected by executing tests on the `.so` library files compiled for Facto-DSA implementations.

- Correctness. Algorithm is validated with KAT or single run of `keygen+sign+verify`.
- Clock cycles for all platforms. Previously commented (x86-64,ARM,MacOS,RISC-V)
- Throughput, operations per second, cycles per operation
- Resource consumption. Measures usage of static memory and peak memory.
- Transmission and storage cost. For Facto-DSA these are measured as bytes/op in the tables.

Only a single CPU core is utilised, with 1000 iterations per operation (keygen, sign, verify). All the runs for the three implementations passed the correctness test. A table is included per CPU tested, with all the indicators given for each target security bit level and implementation version.

The following tables report the NGCC self-assessment outputs for the submitted Facto-DSA implementation versions. Each cell gives, from top to bottom, the mean cycle count per operation, operations per second and bytes per operation. Memory columns are reported in bytes.

AMD Ryzen 7 9700X(x86-64) Clock frequency: 3600MHz. Memory: 32GB RAM. Compiler Version: clang 22.1.6. OS: Arch Linux Kernel 7.0.11

Table 1: Facto-DSA Reference implementation results on AMD Ryzen 7 9700X 8-Core Processor (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	4,141,037 916 43,134	1,490,197 2,545 64	486,432 7,793 64	77,824	77,824
256	56,344,458 68 468,622	6,821,710 556 64	5,279,432 719 64	1,105,920	69,632
512	1,231,324,705 4 5,736,162	55,055,193 69 64	65,324,300 59 64	12,185,600	69,632

Table 2: Facto-DSA Performance optimisation version results on AMD Ryzen 7 9700X 8-Core Processor (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	1,227,572 3,089 43,134	771,066 4,918 64	342,838 11,055 64	315,392	143,360
256	15,414,735 247 468,622	2,872,353 1,321 64	3,831,316 990 64	2,990,080	167,936
512	291,948,889 13 5,736,162	19,391,024 196 64	47,504,840 80 64	35,987,456	389,120

Table 3: Facto-DSA Resource optimisation version results on AMD Ryzen 7 9700X 8-Core Processor (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	6,077,342	3,753,330	400,654	40,960	40,960
	625	1,011	9,460		
	43,134	64	64		
256	87,490,060	23,154,377	4,389,826	516,096	40,960
	44	164	864		
	468,622	64	64		
512	2,282,596,240	222,074,632	54,969,829	6,004,736	40,960
	1.6616	18	69		
	5,736,162	64	64		

AMD EPYC 7413 (x86-64) Clock frequency: 2650MHz. Memory: 32GB RAM. Compiler Version: clang 22.1.6. OS: Arch Linux Kernel 7.0.11

Table 4: Facto-DSA Reference implementation results on AMD EPYC 7413 (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	5,246,837 506 43,134	1,489,816 1,779 64	544,615 4,863 64	77,824	77,824
256	69,850,385 38 468,622	7,244,582 366 64	6,038,055 439 64	1,105,920	69,632
512	1,538,985,104 1.7219 5,736,162	57,673,065 46 64	73,747,235 36 64	12,185,600	69,632

Table 5: Facto-DSA Performance optimisation version results on AMD EPYC 7413 (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	1,647,062 1,609 43,134	931,452 2,844 64	420,228 6,302 64	307,200	135,168
256	21,073,893 126 468,622	3,368,975 787 64	4,711,235 563 64	2,990,080	167,936
512	398,461,718 7 5,736,162	21,363,696 125 64	58,178,480 46 64	35,983,360	385,024

Table 6: Facto-DSA Resource optimisation version results on AMD EPYC 7413 (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	7,451,711	3,640,131	458,633	40,960	40,960
	356	728	5,774		
	43,134	64	64		
256	107,232,369	22,482,540	5,123,797	516,096	40,960
	25	118	518		
	468,622	64	64		
512	2,592,306,764	233,816,749	64,923,022	6,004,736	40,960
	1.0223	12	41		
	5,736,162	64	64		

XEON E3-1220 (x86-64) Clock frequency: 3100MHz. Memory: 4GB RAM. Compiler Version: clang 21.1.8. OS: Debian Kernel 7.0.12

Table 7: Facto-DSA Reference implementation results on XEON E3-1220 (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	7,664,892	2,440,880	857,300	73,728	73,728
	314	983	2,798		
	43,134	64	64		
256	104,970,821	13,171,202	9,620,561	1,105,920	69,632
	23	183	250		
	468,622	64	64		
512	2,313,589,940	105,287,185	127,873,255	12,185,600	69,632
	1.0373	23	19		
	5,736,162	64	64		

Table 8: Facto-DSA Performance optimisation version results on XEON E3-1220 (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	3,024,043	1,308,737	742,217	327,680	155,648
	794	1,833	3,231		
	43,134	64	64		
256	38,474,165	6,782,550	8,249,714	2,981,888	159,744
	63	354	291		
	468,622	64	64		
512	831,912,710	48,059,344	104,008,777	35,966,976	368,640
	3	50	24		
	5,736,162	64	64		

Table 9: Facto-DSA Resource optimisation version results on XEON E3-1220 (x86-64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	15,398,091 156 43,134	4,788,098 502 64	741,083 3,236 64	40,960	40,960
256	229,092,486 11 468,622	30,429,514 79 64	8,757,023 275 64	516,096	40,960
512	5,891,107,244 0.4074 5,736,162	290,332,127 9 64	106,892,710 23 64	6,004,736	40,960

Graviton3 (aarch64) Clock frequency: 2600MHz. Memory: 4GB RAM. Compiler Version: clang 15.0.7. OS: Amazon Linux Kernel 6.1.174

Table 10: Facto-DSA Reference implementation results on Graviton3 (aarch64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	4,406,447	1,047,361	592,769	270,336	270,336
	589	2,469	4,349		
	43,134	64	64		
256	59,513,064	5,007,281	6,667,373	1,302,528	270,336
	44	519	390		
	468,622	64	64		
512	1,285,741,640	41,499,179	83,475,324	12,386,304	270,336
	3	63	32		
	5,736,162	64	64		

Table 11: Facto-DSA Performance optimisation version results on Graviton3 (aarch64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	2,109,364	1,030,936	554,070	450,560	278,528
	1,229	2,509	4,650		
	43,134	64	64		
256	28,446,411	4,836,410	6,250,486	3,076,096	307,200
	92	537	416		
	468,622	64	64		
512	587,453,732	27,610,468	77,641,134	36,134,912	536,576
	5	95	34		
	5,736,162	64	64		

Table 12: Facto-DSA Resource optimisation version results on Graviton3 (aarch64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	8,263,380	1,637,837	547,490	266,240	266,240
	315	1,582	4,706		
	43,134	64	64		
256	125,759,238	8,897,697	6,117,433	737,280	266,240
	21	292	425		
	468,622	64	64		
512	3,381,150,543	83,137,107	75,815,244	6,225,920	266,240
	0.7682	32	35		
	5,736,162	64	64		

Cortex-A72 (aarch64) Clock frequency: 1500MHz. Memory: 4GB RAM. Compiler Version: clang 19.1.7. OS: Linux Raspberrypi Kernel 6.18.34

Table 13: Facto-DSA Reference implementation results on Cortex-A72 (aarch64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	10,131,062	2,346,053	1,211,694	204,800	139,264
	178	767	1,485		
	43,134	64	64		
256	142,865,721	12,512,616	14,054,664	1,171,456	139,264
	13	144	129		
	468,622	64	64		
512	3,057,669,651	105,005,872	176,929,052	12,255,232	139,264
	0.5887	18	11		
	5,736,162	64	64		

Table 14: Facto-DSA Performance optimisation version results on Cortex-A72 (aarch64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	4,820,965	2,301,417	1,042,601	319,488	147,456
	374	782	1,726		
	43,134	64	64		
256	69,343,785	11,735,263	11,505,712	2,945,024	176,128
	26	154	157		
	468,622	64	64		
512	1,462,090,761	93,618,336	142,839,303	36,003,840	405,504
	1.2311	20	13		
	5,736,162	64	64		

Table 15: Facto-DSA Resource optimisation version results on Cortex-A72 (aarch64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	21,057,435 86 43,134	2,868,402 628 64	1,033,262 1,741 64	200,704	135,168
256	343,448,671 6 468,622	17,349,381 104 64	11,629,630 155 64	606,208	135,168
512	10,130,069,391 0.1777 5,736,162	170,870,870 11 64	144,095,237 13 64	6,094,848	135,168

Apple MacBook Pro (arm64) Clock frequency: 4500Mhz. Memory: 24GB. Compiler Version: clang 17.0.0. OS: Darwin Kernel 25.5.0

Remark For Apple OS tests, Static and Peak memory values were not collected as nggc.bench program has non POSIX code to measure these. It is not adapted for other non Linux OS.

Table 16: Facto-DSA Reference implementation results on Apple MacBook Pro (arm64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op
128	3,195,864	1,059,581	499,817
	1,258	4,067	8,184
	43,134	64	64
256	42,857,758	4,424,497	5,681,989
	95	961	723
	468,622	64	64
512	940,562,848	26,883,320	71,383,028
	4	149	56
	5,736,162	64	64

Table 17: Facto-DSA Performance optimisation version results on Apple MacBook Pro (arm64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op
128	1,515,910	1,094,377	450,346
	2,524	3,800	8,951
	43,134	64	64
256	18,916,741	4,448,331	5,173,002
	213	925	781
	468,622	64	64
512	413,893,745	23,993,452	63,070,404
	10	169	64
	5,736,162	64	64

Table 18: Facto-DSA Resource optimisation version results on Apple MacBook Pro (arm64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op
128	4,883,469	1,182,778	461,235
	813	3,345	8,692
	43,134	64	64
256	72,210,203	6,737,384	5,285,602
	56	597	768
	468,622	64	64
512	1,939,567,041	60,054,927	64,517,180
	2	67	63
	5,736,162	64	64

Ky X1 RISCv (riscv64) Clock frequency: 1600MHz. Memory: 8GB RAM. Compiler Version: clang 18.1.3. OS: Orange Pi RV2 Linux Kernel 6.6.63, Ubuntu 24.04 Based

Table 19: Facto-DSA Reference implementation results on Ky(R) X1 processor (riscv64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	45,788,599 35 43,134	9,884,838 162 64	3,113,792 511 64	40,960	40,960
256	636,986,678 3 468,622	80,930,042.2 19.762 64	36,721,048 44 64	1,077,248	45,056
512	14,737,213,194 0.109 5,736,162	702,352,670 3 64	469,500,662 4 64	12,156,928	40,960

Table 20: Facto-DSA Performance optimisation version results on Ky(R) X1 processor (riscv64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	16,336,692 98 43,134	10,560,394 151 64	1,995,124 795 64	249,856	77,824
256	243,501,850 6 468,622	57,850,264 28 64	22,210,538 72 64	2,875,392	106,496
512	5,672,315,024 0.282 5,736,162	1,091,160,221 1.466 64	273,707,443 6 64	35,934,208	335,872

Table 21: Facto-DSA Resource optimisation version results on Ky(R) X1 processor (riscv64). Operation cells list cycles/op, ops/s and bytes/op.

Target bits	KeyGen cycles/op ops/s bytes/op	Sign cycles/op ops/s bytes/op	Verify cycles/op ops/s bytes/op	Peak memory bytes	Static memory bytes
128	75,921,225 21 43,134	7,981,283 200 64	2,638,885 602 64	32,768	32,768
256	1,449,446,070 1.104 468,622	118,116,932 14 64	29,519,500 54 64	503,808	32,768
512	49,032,343,824 0.033 5,736,162	725,065,648 2 64	362,313,097 5 64	5,992,448	32,768